

J.P. DAWER INSTITUTE OF INFORMATION AND
COMMUNICATION TECHNOLOGY

Academic Year: 2026 - 27

304 – Open Source Web Development

Practical Assignment - I

➤ Submitted by:

Name: Thakkar Disha Vipulbhai

Class: ICT-3

Rollno.: 54

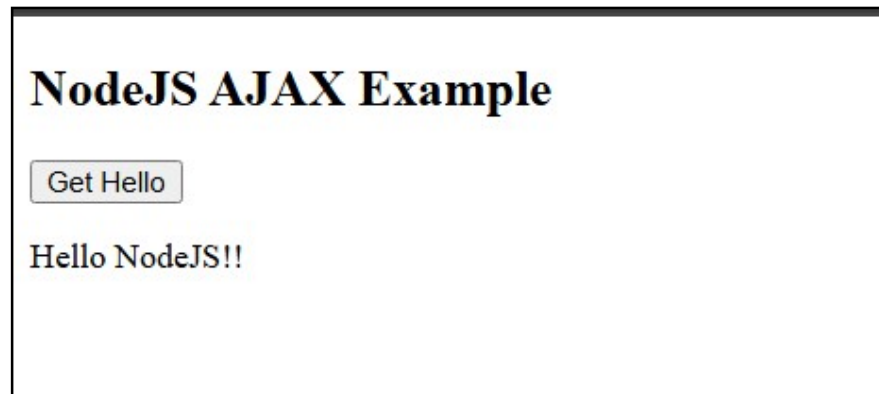
➤ Submitted to:

Prof. Rupal Panchal

Que.1 : Develop nodejs application with following requirements:

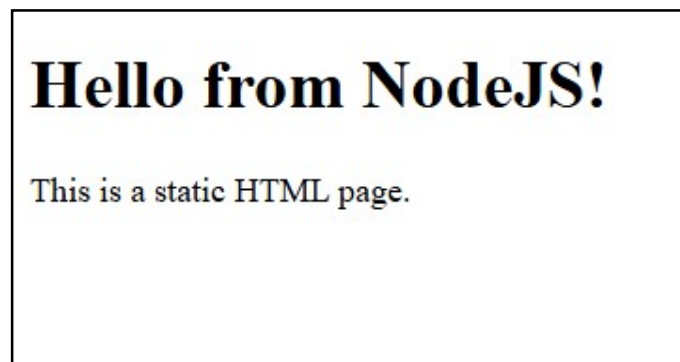
- Develop a route "/gethello" with GET method. It displays "Hello NodeJS!!" as response.
- Make an HTML page and display.
- Call "/gethello" route from HTML page using AJAX call. (Any frontend AJAX call API can be used.)

Output :



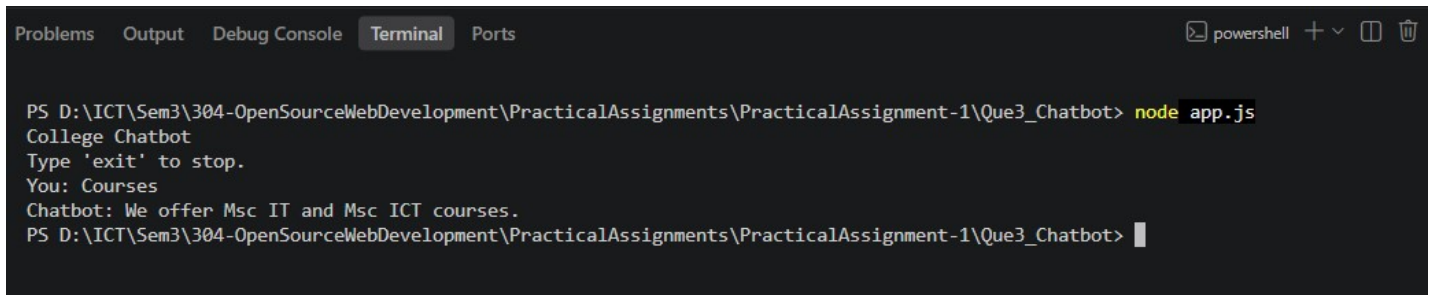
Que.2: Develop a web server which serves static resources.

Output :



Que.3: Develop a module for domain specific chatbot and use it in a command line application.

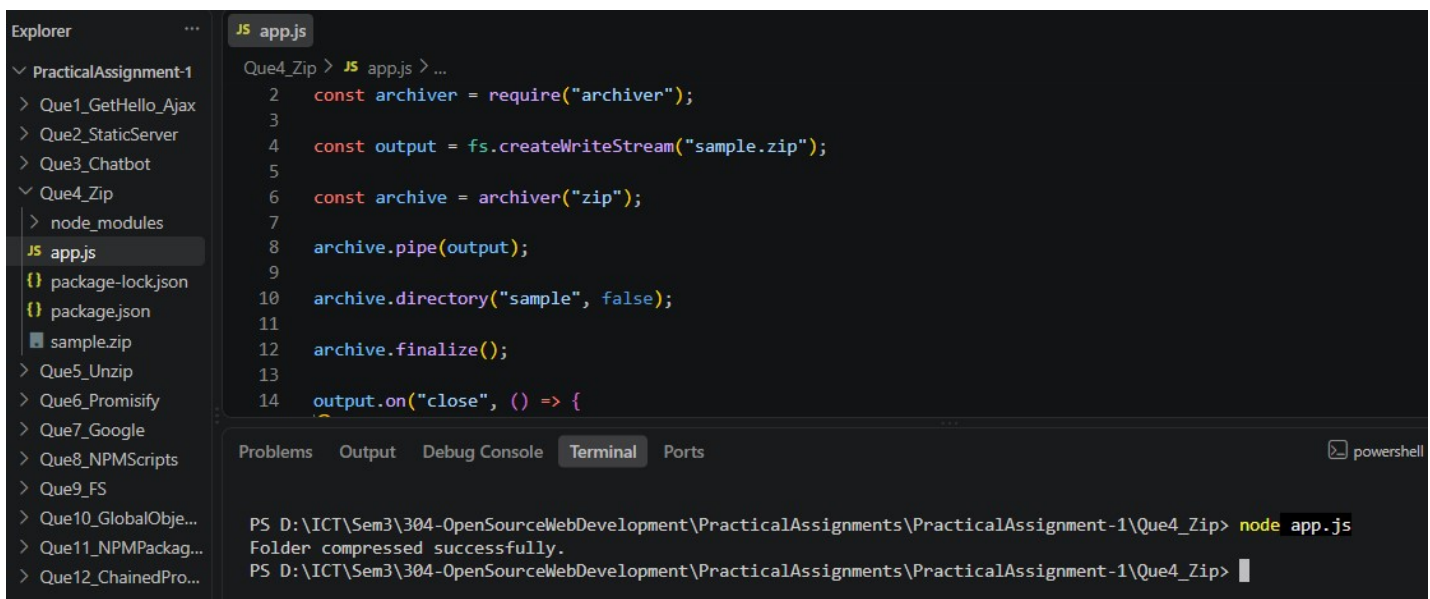
Output :



```
Problems Output Debug Console Terminal Ports
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que3_Chatbot> node app.js
College Chatbot
Type 'exit' to stop.
You: Courses
Chatbot: We offer Msc IT and Msc ICT courses.
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que3_Chatbot>
```

Que.4: Write a program to create a compressed zip file for a folder.

Output :



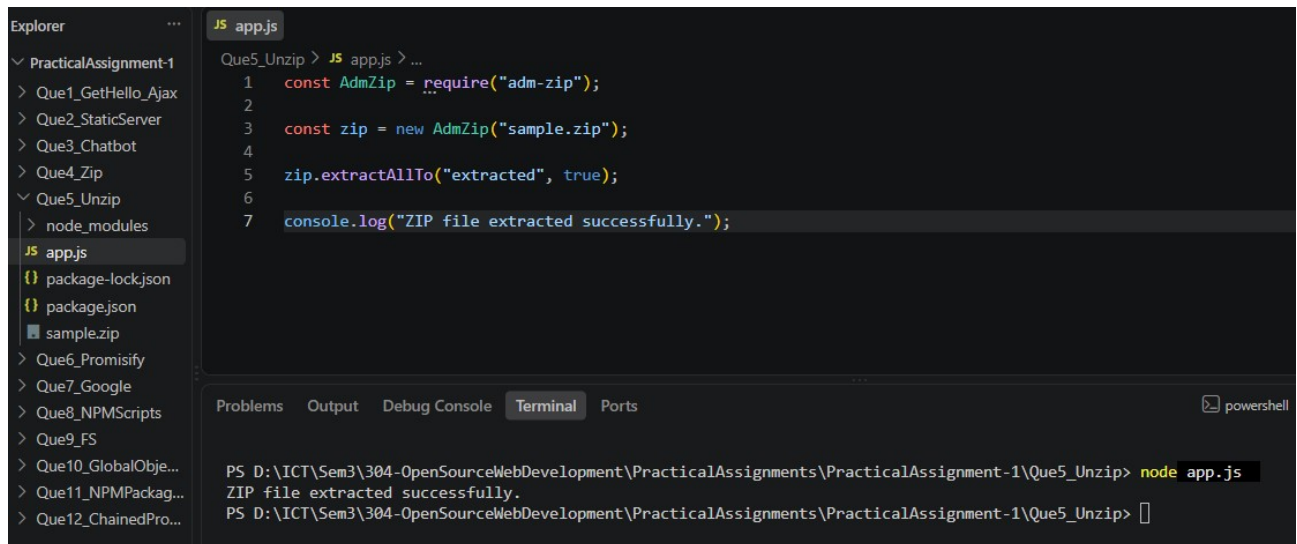
```
Explorer
PracticalAssignment-1
  > Que1_GetHello_Ajax
  > Que2_StaticServer
  > Que3_Chatbot
  > Que4_Zip
    > node_modules
    JS app.js
    {} package-lock.json
    {} package.json
    sample.zip
  > Que5_Unzip
  > Que6_Promisify
  > Que7_Google
  > Que8_NPMScripts
  > Que9_FS
  > Que10_GlobalObje...
  > Que11_NPMPackag...
  > Que12_ChainedPro...

JS app.js
Que4_Zip > JS app.js > ...
2   const archiver = require("archiver");
3
4   const output = fs.createWriteStream("sample.zip");
5
6   const archive = archiver("zip");
7
8   archive.pipe(output);
9
10  archive.directory("sample", false);
11
12  archive.finalize();
13
14  output.on("close", () => {
    ...

Problems Output Debug Console Terminal Ports
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que4_Zip> node app.js
Folder compressed successfully.
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que4_Zip>
```

Que.5: Write a program to extract a zip file.

Output :



The screenshot shows a VS Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with folders 'Que1_GetHello_Ajax', 'Que2_StaticServer', 'Que3_Chatbot', 'Que4_Zip', 'Que5_Unzip', and 'node_modules'. The 'Que5_Unzip' folder is selected, showing files 'app.js', 'package-lock.json', 'package.json', and 'sample.zip'. The code editor shows the content of 'app.js' with the following code:

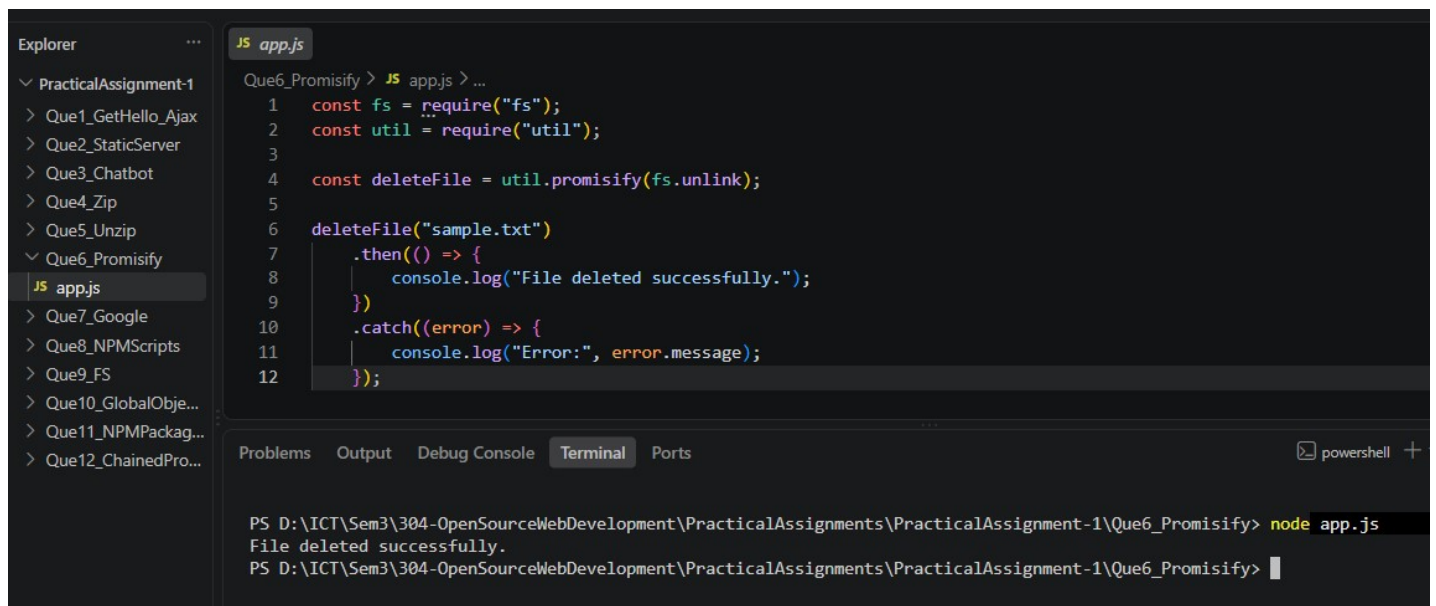
```
1 const AdmZip = require("adm-zip");
2
3 const zip = new AdmZip("sample.zip");
4
5 zip.extractAllTo("extracted", true);
6
7 console.log("ZIP file extracted successfully.");
```

Below the code editor is a terminal window with the following output:

```
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que5_Unzip> node app.js
ZIP file extracted successfully.
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que5_Unzip>
```

Que.6: Write a program to promisify fs.unlink function and call it.

Output :



The screenshot shows a VS Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with folders 'Que1_GetHello_Ajax', 'Que2_StaticServer', 'Que3_Chatbot', 'Que4_Zip', 'Que5_Unzip', 'Que6_Promisify', and 'node_modules'. The 'Que6_Promisify' folder is selected, showing files 'app.js', 'package-lock.json', 'package.json', and 'sample.txt'. The code editor shows the content of 'app.js' with the following code:

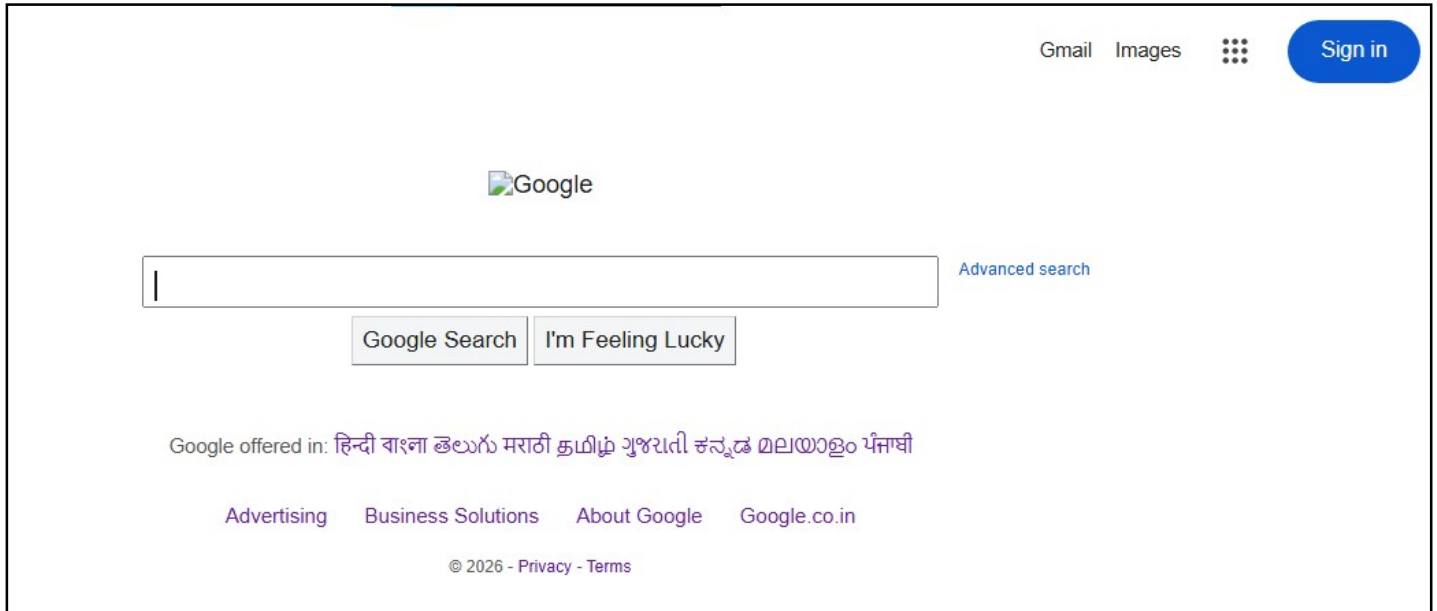
```
1 const fs = require("fs");
2 const util = require("util");
3
4 const deleteFile = util.promisify(fs.unlink);
5
6 deleteFile("sample.txt")
7   .then(() => {
8     console.log("File deleted successfully.");
9   })
10  .catch((error) => {
11    console.log("Error:", error.message);
12  });
```

Below the code editor is a terminal window with the following output:

```
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que6_Promisify> node app.js
File deleted successfully.
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que6_Promisify>
```

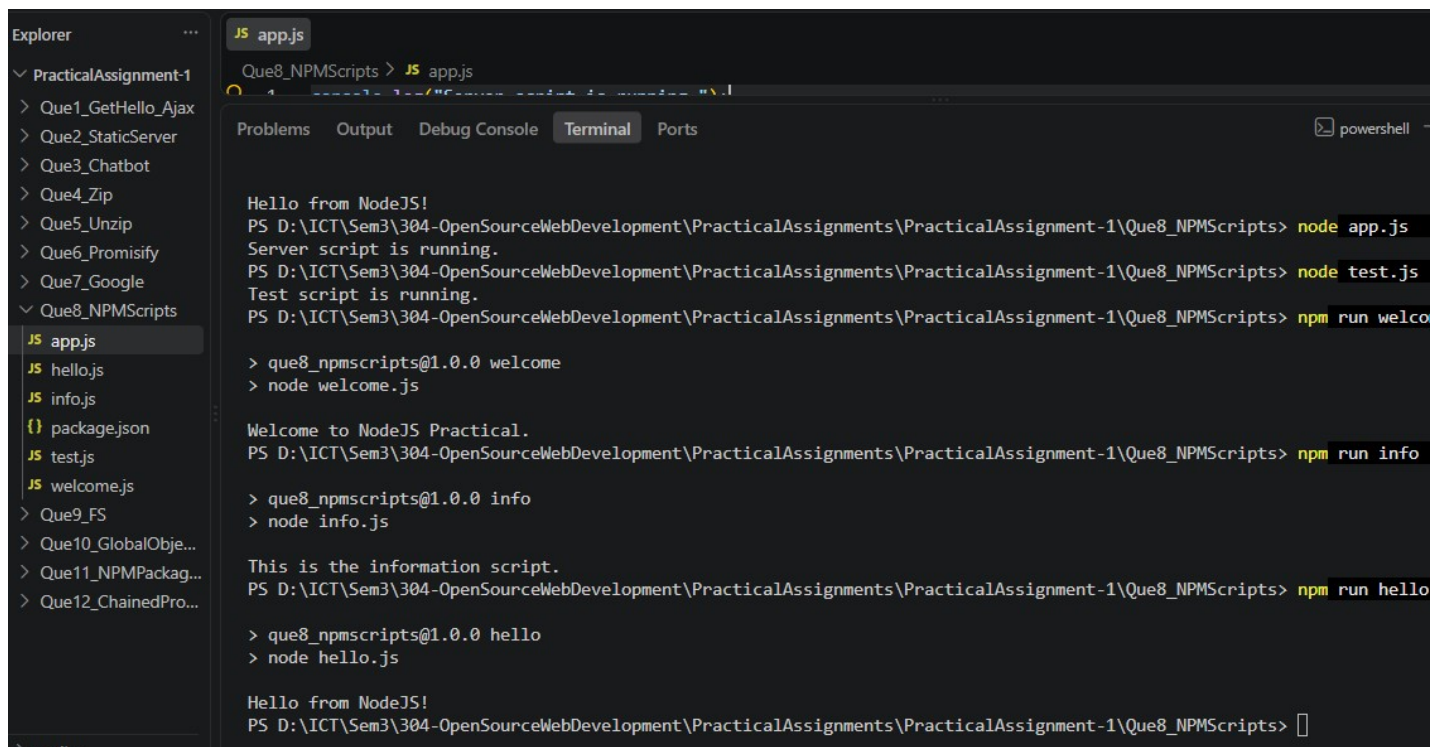
Que.7: Develop a node web server with a route /google which can fetch data of google page using fetch() using async-await model and show as a response.

Output :



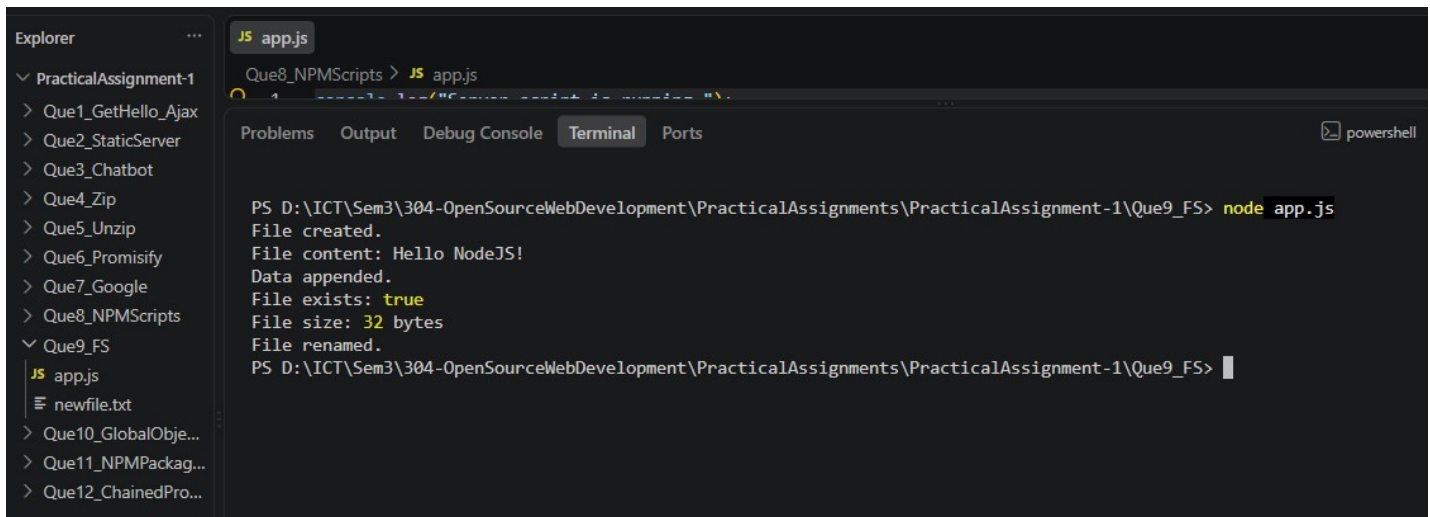
Que.8: Set a server script, a test script and 3 user defined scripts in package.json file in your nodejs application.

Output :



Que.9: A program which calls useful functions in fs module.

Output :

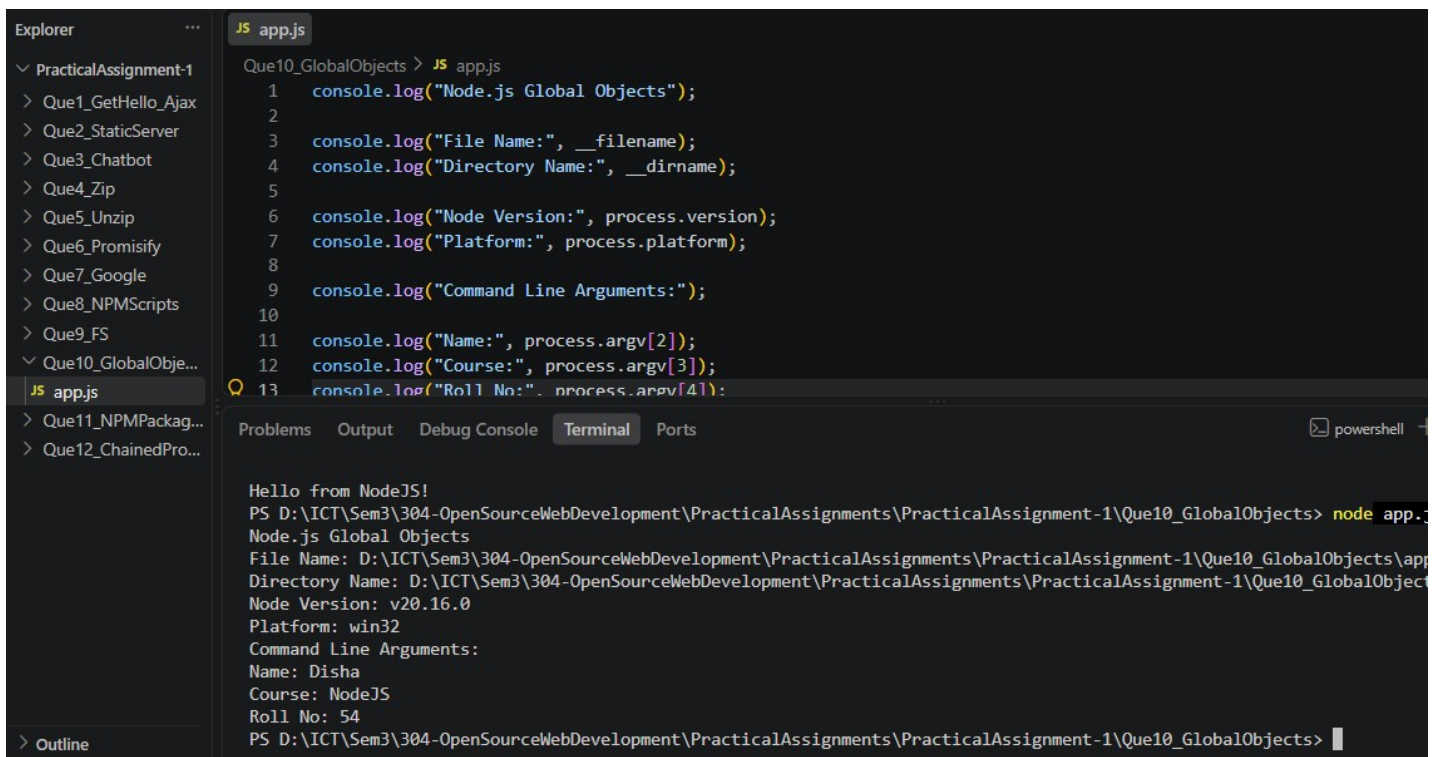


```
JS app.js
Que8_NPMScripts > JS app.js

Problems Output Debug Console Terminal Ports
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que9_FS> node app.js
File created.
File content: Hello NodeJS!
Data appended.
File exists: true
File size: 32 bytes
File renamed.
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que9_FS>
```

Que.10: A program which uses global objects in nodejs. Also read command line arguments and print it.

Output :



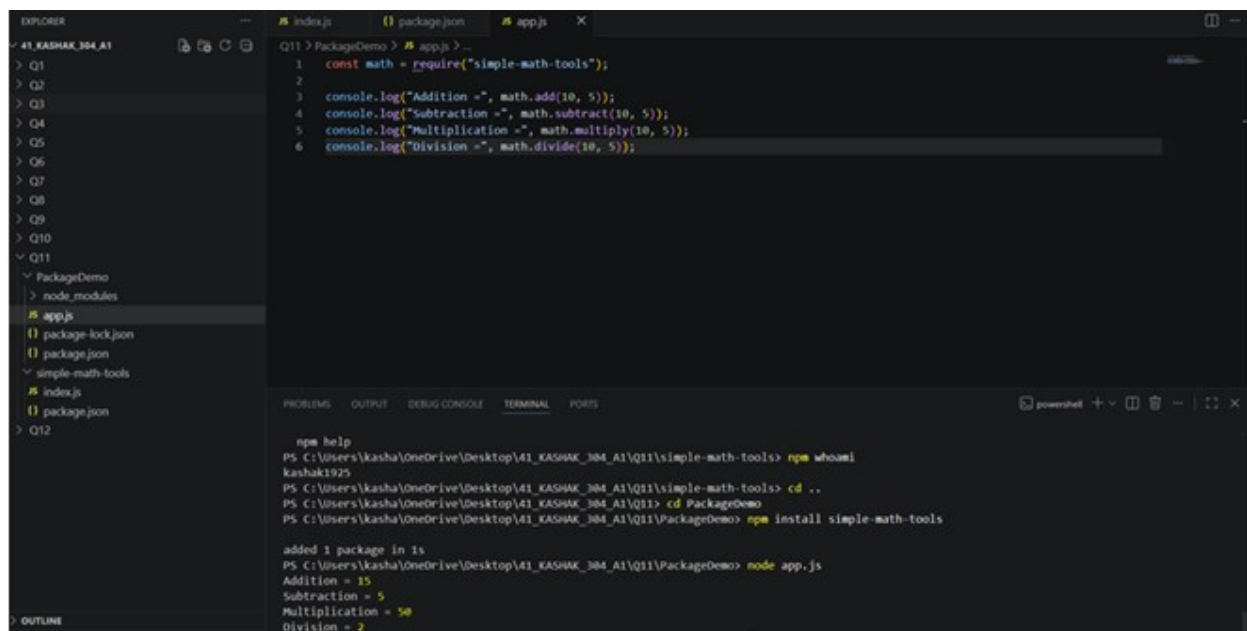
```
JS app.js
Que10_GlobalObjects > JS app.js

1 console.log("Node.js Global Objects");
2
3 console.log("File Name:", __filename);
4 console.log("Directory Name:", __dirname);
5
6 console.log("Node Version:", process.version);
7 console.log("Platform:", process.platform);
8
9 console.log("Command Line Arguments:");
10
11 console.log("Name:", process.argv[2]);
12 console.log("Course:", process.argv[3]);
13 console.log("Roll No:", process.argv[4]);

Problems Output Debug Console Terminal Ports
Hello from NodeJS!
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que10_GlobalObjects> node app.js
Node.js Global Objects
File Name: D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que10_GlobalObjects\app.js
Directory Name: D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que10_GlobalObjects
Node Version: v20.16.0
Platform: win32
Command Line Arguments:
Name: Disha
Course: NodeJS
Roll No: 54
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que10_GlobalObjects>
```


Que.11 Develop a useful package and publish it on npmjs.com. Write a program to use it.

Output :



The screenshot shows a VS Code editor with a file explorer on the left and a terminal at the bottom. The file explorer shows a project structure for 'Q11' with a 'PackageDemo' folder containing 'app.js', 'package-lock.json', and 'package.json'. The 'app.js' file is open in the editor, showing code that uses the 'simple-math-tools' package for addition, subtraction, multiplication, and division. The terminal shows the command 'npm install simple-math-tools' being executed, followed by 'node app.js', which outputs the results of the calculations.

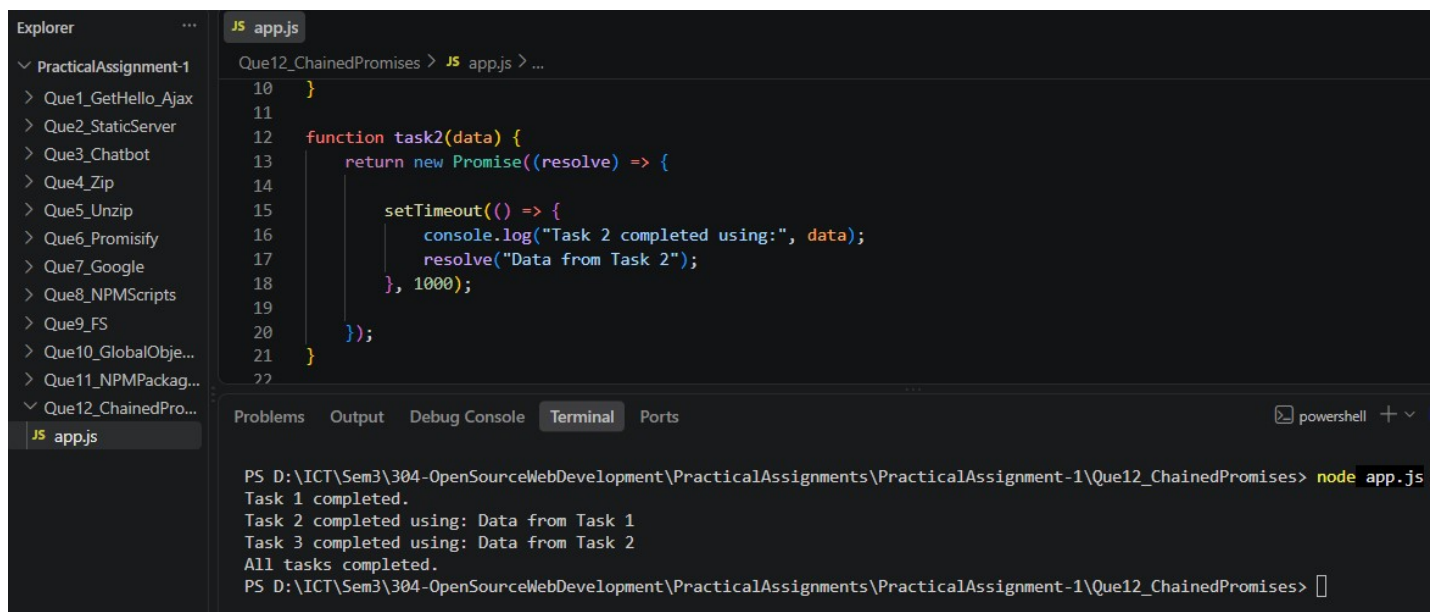
```
Q11 > PackageDemo > JS app.js > ...
1  const math = require("simple-math-tools");
2
3  console.log("Addition =", math.add(10, 5));
4  console.log("Subtraction =", math.subtract(10, 5));
5  console.log("Multiplication =", math.multiply(10, 5));
6  console.log("Division =", math.divide(10, 5));

npm help
PS C:\Users\kasha\OneDrive\Desktop\41_KASHAK_304_A1\Q11\simple-math-tools> npm whoami
kashak1925
PS C:\Users\kasha\OneDrive\Desktop\41_KASHAK_304_A1\Q11\simple-math-tools> cd ..
PS C:\Users\kasha\OneDrive\Desktop\41_KASHAK_304_A1\Q11> cd PackageDemo
PS C:\Users\kasha\OneDrive\Desktop\41_KASHAK_304_A1\Q11\PackageDemo> npm install simple-math-tools

added 1 package in 1s
PS C:\Users\kasha\OneDrive\Desktop\41_KASHAK_304_A1\Q11\PackageDemo> node app.js
Addition = 15
Subtraction = 5
Multiplication = 50
Division = 2
```

Que.12: Write a node program having at least 3 dependent tasks and call using chained promises.

Output :



The screenshot shows a VS Code editor with a file explorer on the left and a terminal at the bottom. The file explorer shows a project structure for 'PracticalAssignment-1' with a 'Que12_ChainedPromises' folder containing 'app.js'. The 'app.js' file is open in the editor, showing code that uses chained promises to execute three tasks sequentially. The terminal shows the command 'node app.js' being executed, followed by the output of the tasks.

```
Que12_ChainedPromises > JS app.js > ...
10  }
11
12  function task2(data) {
13    return new Promise((resolve) => {
14      setTimeout(() => {
15        console.log("Task 2 completed using:", data);
16        resolve("Data from Task 2");
17      }, 1000);
18    });
19  }
20
21  }
22

PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que12_ChainedPromises> node app.js
Task 1 completed.
Task 2 completed using: Data from Task 1
Task 3 completed using: Data from Task 2
All tasks completed.
PS D:\ICT\Sem3\304-OpenSourceWebDevelopment\PracticalAssignments\PracticalAssignment-1\Que12_ChainedPromises>
```